

Mohammad Rajabi

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Summary

AI Engineer with 2 years of experience across AI and software engineering, focused on designing and building production-grade agentic systems, RAG pipelines, LLM integrations, and NLP solutions. Master's in Computer Science with specialisation in AI. Holder of a Dutch Orientation Year (Zoekjaar) visa, available from June 2026.

Work Experience

AI Engineer

July 2025 – Present

Pet 24-7 Global Ltd

London, United Kingdom

 Live app: App Store – Google Play

Joined as the first AI engineer; built the full AI stack end-to-end — now live with real users across the UK, US, and India and actively expanding.

Agentic AI System

- Designed and built a **LangGraph ReAct agent** with pluggable tools — knowledge base retrieval, semantic history search, web search, image generation, multimodal OCR, speech-to-text, and PDF generation — enabling autonomous multi-step AI consultations.
- Engineered a **hybrid memory architecture** combining PostgreSQL session cache with Milvus vector search for fast recent-turn recall and long-term semantic history retrieval.
- Built **MCP servers** to expose agent tools as modular, standardized interfaces.
- Deployed on **AWS** with CI/CD pipelines and integrated **Langfuse** for end-to-end LLM observability across all agent steps and tool calls.
- Benchmarked **Milvus**, **Qdrant**, and **Weaviate** across chunking strategies (recursive, semantic, fixed-size), evaluating latency, throughput, and retrieval quality to drive architecture decisions.

Production RAG Chatbot

- Built and shipped the first production RAG chatbot serving real users from day one, applying advanced retrieval techniques including **multi-step query decomposition**, **step-back prompting**, and **cross-encoder re-ranking** over a domain-specific knowledge base.
- Applied advanced **prompt engineering** techniques for retrieval optimization and response quality improvement, with multilingual query handling for international users.
- Designed a **per-user episodic memory system** using vector embeddings to semantically retrieve past interactions, enabling context-aware follow-up responses across sessions.
- Delivered **real-time SSE streaming** and a multimodal pipeline covering OCR, image, video, voice, and PDF processing.

NLP Analytics Dashboard

- Built an end-to-end **NLP analytics pipeline** using a two-layer NER system and **BERTopic-based** topic modeling with LLM-generated labels.
- Developed a **real-time analytics dashboard** surfacing entity trends, topic modeling, co-occurrence analysis, and regional breakdowns via a React + FastAPI backend.

Full Stack Developer

Nov 2023 – Nov 2024

MeNEW - Personal Digital Menu

Part-time

Digital Menu Platform

- Built and launched a production-grade digital menu platform using **Next.js (TypeScript)**, **Django REST API**, and **GraphQL**, serving real restaurant customers across multiple locations.
- Owned end-to-end delivery — from **API design and database architecture** to **Dockerized deployments** and **CI/CD pipelines** — collaborating in an Agile team with regular code reviews and sprint cycles.

Technical Skills

Languages	Python, Java, C++ , JavaScript, TypeScript
AI/ML	NLP, LLM, RAG, Agentic AI, Deep Learning, Computer Vision, Classification, Regression, Clustering
Frameworks	PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, FastAPI, Keras, Spark
Libraries	Pandas, NumPy, NLTK, SciPy, BERTopic, Matplotlib, Plotly, Seaborn
Vector Databases	Milvus, Qdrant, Weaviate, ChromaDB
Cloud & Storage	AWS S3, Google Cloud (Vertex AI)
LLM APIs	Google Gemini, OpenAI API, MCP
Web & DevOps	React, Next.js, Node.js, Django, Docker, Git, SQL, MongoDB, PostgreSQL, Prisma, JWT

Education

University of Padua	Padova, Italy
Master in Computer Science, Specialisation in Artificial Intelligence	Sep 2022 – Sep 2025
GPA: 105/110	
Thesis: Improving Navigation in LLM-Based AI Frameworks	October 2024 – July 2025
Designed and built an agentic AI framework where an LLM acts as the reasoning engine, dynamically selecting and composing domain-specific tools at runtime via structured APIs and in-context prompting. The system follows a ReAct-style reasoning loop — generating executable code, observing intermediate results, and iterating toward a solution. Benchmarked against end-to-end and modular baselines to evaluate performance and robustness.	
Supervisor: Prof. Lamberto Ballan	

Course Projects

- **Sentiment Analysis and Topic Modeling with Fine-tuned Transformer Models** — Fine-tuned encoder-based Transformers and LSTM/GRU architectures on real-world data; applied BERTopic and LDA for latent theme discovery linked to sentiment patterns. (GitHub)
- **CNN and Vision Transformer Fine-tuning for Object Detection** — Fine-tuned CNN and ViT architectures on PASCAL VOC; extended with OWL-ViT2 for zero-shot detection, highlighting trade-offs between supervised accuracy and open-world flexibility. (GitHub)
- **LLM Pruning and Fine-Tuning for Efficient Text Summarization** — Applied SparseGPT pruning and LoRA fine-tuning to compress and adapt LLMs with minimal performance loss, enabling deployment on resource-constrained hardware.

For other works, explore more on my GitHub

Languages

English: C2	Dutch: A2	Italian: A2	Persian: Native
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